

Perfect Secrecy in the Wild

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IN HUMBLE SUBSTITUTION FOR ALKIS

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Q1: When is Alice's problem feasible?

Q2: More generally, what information about X can be revealed?

Part I — full information

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In our setting, $P_{Y|X}$ is given as a primitive.

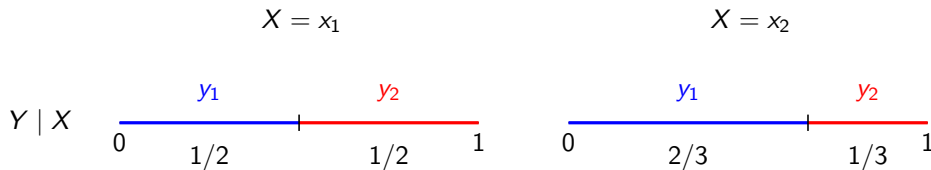
Y may not be uniform and independent of X .

The characterization

Theorem 1: There exist a random variable Z and a finitely supported joint distribution Q_{XYZ} satisfying informativeness, perfect secrecy and consistency *if and only if*

$$\text{for all } y \in \text{supp } P_Y, \quad \sum_{x \in \text{supp } P_X} P_{Y|X}(y | x) \leq 1.$$

Illustration of necessity



Representation of signals as partitions of $\{x_1, x_2\} \times [0, 1]$. (Green and Stokey, 1978)

Length of intervals represent $P_{Y|X}(y | x)$.

An independent uniform draw from $[0, 1]$ determines Y conditional on X .

Illustration of necessity

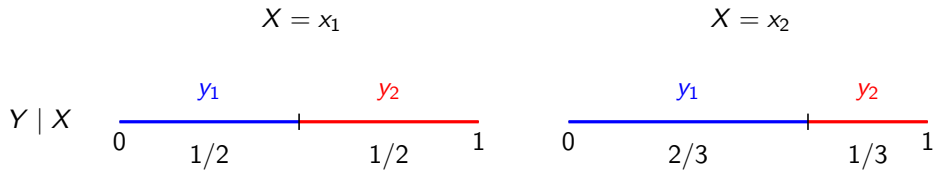
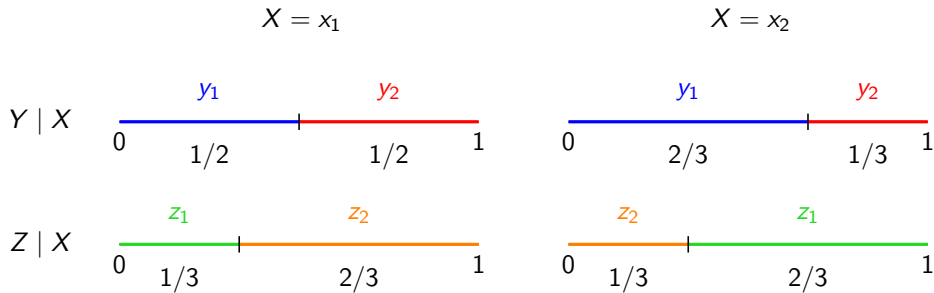
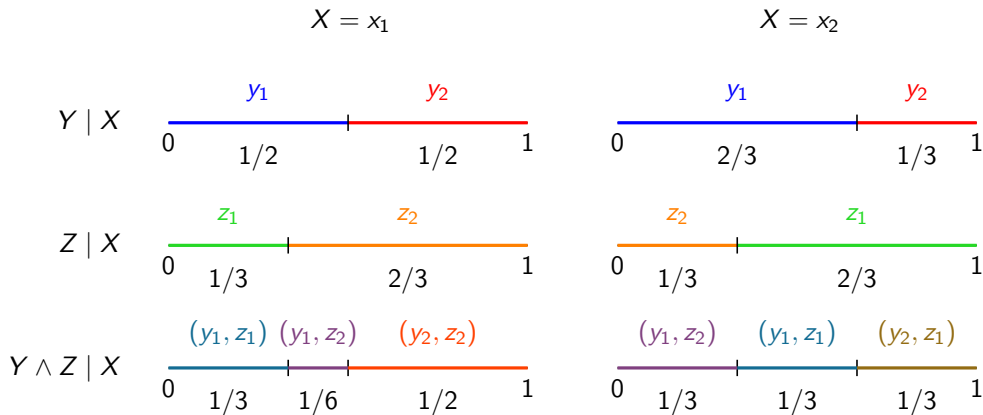


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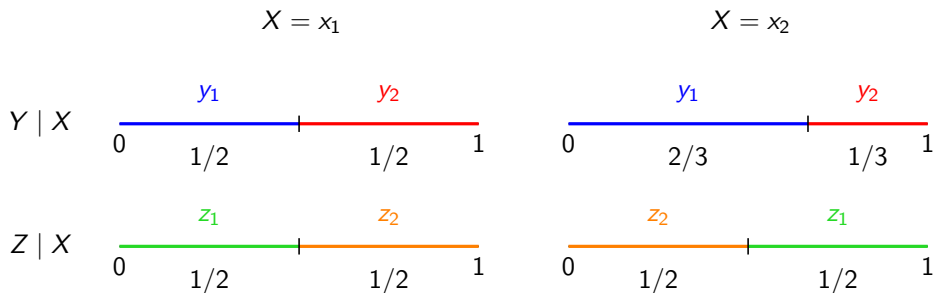
Allows to model arbitrary dependence between signals.

Illustration of necessity



The join of two signals is their coarsest common refinement.

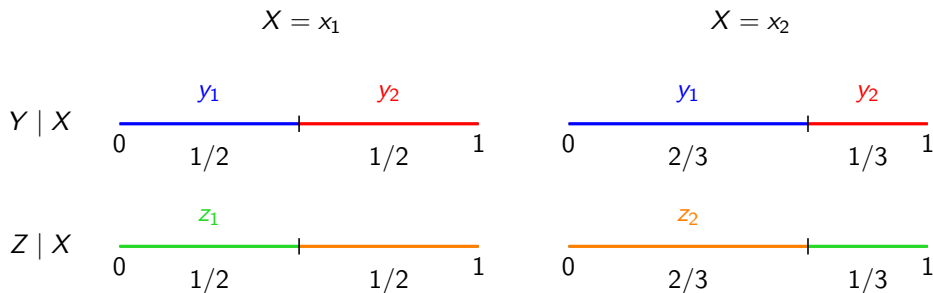
Illustration of necessity



For Z to be perfectly secret it must hold $Q_{Z|X}(\cdot | x_1) = Q_{Z|X}(\cdot | x_2)$

That is, each z must occur at x_1 with the same mass it occurs at x_2 .

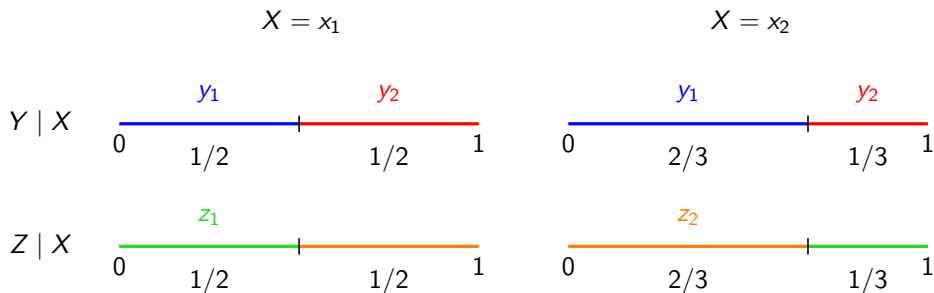
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Say we want to reveal X conditional on $Y = y_1$ (which fails the condition).

Then, z_1 must appear under y_1 in x_1 but not under y_1 in x_2 .

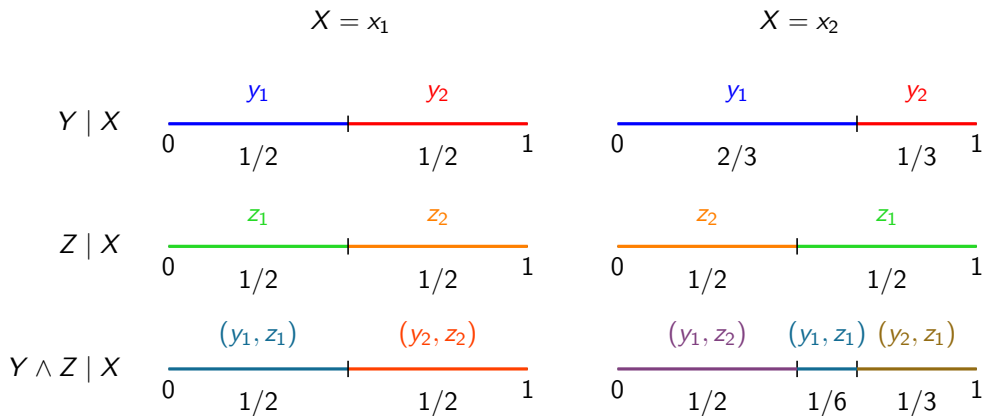
Illustration of necessity



To reveal X conditional on $Y = y_1$ under perfect secrecy,
we must be able to fit the mass of y_1 in x_1 under y_2 in x_2 .

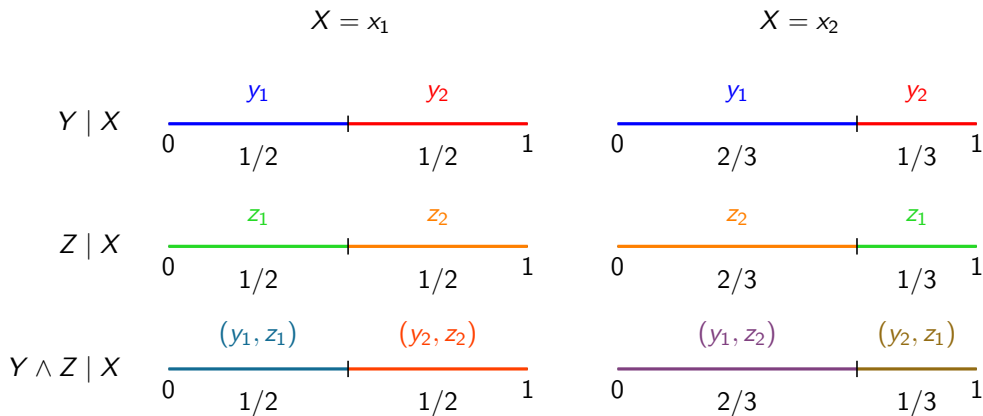
Requires $P_{Y|X}(y_1 | x_1) \leq P_{Y|X}(y_2 | x_2) = 1 - P_{Y|X}(y_1 | x_2)$.

Illustration of necessity



So, any Z will either fail informativeness,

Illustration of necessity



or it will fail perfect secrecy.

Illustration of sufficiency

$$\text{Suppose } \mathbf{P}_X = \begin{pmatrix} 1/4 \\ 3/4 \end{pmatrix} \begin{matrix} x_1 \\ x_2 \end{matrix}, \quad \text{and} \quad \mathbf{P}_{Y|X} = \begin{pmatrix} \overset{y_1}{1/3} & \overset{y_2}{2/3} \\ 2/3 & 1/3 \end{pmatrix} \begin{matrix} x_1 \\ x_2 \end{matrix}.$$

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By the Birkhoff-von Neumann theorem,

$$\mathbf{P}_{Y|X} = \frac{1}{3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \frac{2}{3} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

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The joint that solves the problem is given by

$$\mathbf{Q}_{XYZ}[\cdot, \cdot, z_1] = \frac{1}{3} \begin{pmatrix} \overset{y_1}{1/4} & \overset{y_2}{0} \\ 0 & 3/4 \end{pmatrix}, \quad \mathbf{Q}_{XYZ}[\cdot, \cdot, z_2] = \frac{2}{3} \begin{pmatrix} \overset{y_1}{0} & \overset{y_2}{1/4} \\ 3/4 & 0 \end{pmatrix}.$$

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Perfect secrecy is satisfied

$$\mathbf{Q}_{XZ}[\cdot, z_1] = \frac{1}{3} \begin{pmatrix} 1/4 \\ 3/4 \end{pmatrix}, \quad \mathbf{Q}_{XZ}[\cdot, z_2] = \frac{2}{3} \begin{pmatrix} 1/4 \\ 3/4 \end{pmatrix}$$

Illustration of sufficiency (cont'd)

Suppose $\mathbf{P}_X = \begin{pmatrix} 1/3 \\ 2/3 \end{pmatrix}$ and $\mathbf{P}_{Y|X} = \begin{pmatrix} 1/2 & 1/3 & 1/6 \\ 1/4 & 1/2 & 1/4 \end{pmatrix}$.

A doubly-stochastic extension of $\mathbf{P}_{Y|X}$ is

$$\mathbf{P}_{Y|X}^{\text{ext}} = \begin{pmatrix} 1/2 & 1/3 & 1/6 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & 1/6 & 7/12 \end{pmatrix}$$

Which decomposes into

$$\mathbf{P}_{Y|X}^{\text{ext}} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \frac{1}{12} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} + \frac{1}{6} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

Ignore the last row and apply the same procedure to get $\mathbf{Q}_{XYZ}$

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Alice can reveal X to Bob under perfect secrecy iff for all $w \in \text{supp } P_W$,

$$\text{for all } y \in \text{supp } P_Y, \quad \sum_{x \in \text{supp } P_X} P_{Y|XW}(y | x, w) \leq 1$$

Part II — implementable information

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In what follows, $\text{supp } P_X = \{1, \dots, n\}$ and $\text{supp } P_Y = \{1, \dots, m\}$.

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The value of $\mathbf{K}$ in decision problem $D = (A, u)$ given a prior P_X is

$$V_{D, P_X}(\mathbf{K}) := \sum_{s=1}^r Q_S(s) \max_{a \in A} \sum_{x=1}^n Q_{X|S}^{\mathbf{K}}(x | s) u(a, x).$$

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A signal $\mathbf{K}$ Blackwell-dominates another $\mathbf{K}'$, written $\mathbf{K} \succsim_B \mathbf{K}'$, if and only if

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Blackwell shows $\mathbf{K} \succsim_B \mathbf{K}'$ iff there exists a row-stochastic $\mathbf{A}$ such that $\mathbf{K}' = \mathbf{K}\mathbf{A}$.

What can Alice say?

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K-informativeness: $Q_{YZ|X} \sim_B \mathbf{K}$

Perfect Secrecy: $Q_{X|Z}(\cdot | z) = P_X$ for all $z \in \text{supp } Q_Z$

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Implementable signals — a characterization

Given an $n \times r$ signal $\mathbf{K}$ with no proportional columns, (with associated r.v. S)

define $L(\mathbf{K})$ to be the set of matrices $\mathbf{\Pi} \in \mathbb{R}_+^{r \times m}$ such that

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Theorem 2: A signal $\mathbf{K}$ is implementable *if and only if* there exists a row-stochastic matrix $\mathbf{\Theta}$ in the convex-hull of $L(\mathbf{K})$ such that

$$\mathbf{P}_{Y|X} = \mathbf{K}\mathbf{\Theta}$$

Implementable signals — a characterization (cont'd)

$\mathbf{K}$ is implementable iff $\mathbf{P}_{Y|X}$ can be obtained as a structured garbling of $\mathbf{K}$.

(a) says that each $\Pi^{(z)}$ is an injection from Y -values to S -values.

so Bob, knowing (Y, Z) , can recover the information on S .

$$Y, Z \xrightarrow{\Pi^{(z)}} S$$

(b) says that each $\mathbf{K}\Pi^{(z)}$ is a well-defined distribution on Y conditional on X .

so Alice can randomize Z independently of X while preserving $P_{Y|X}$

$$X \xrightarrow{\mathbf{K}} S \xrightarrow{\Pi^{(z)}} Y$$

Part III — toward the Blackwell frontier

Unimprovable P_{XY}

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Proposition 1: $P_{Y|X}$ is unimprovable if and only if for every $x \in \text{supp } P_X$,

either $\text{supp } P_{Y|X}(\cdot | x)$ is a singleton,

or $\forall x' \neq x, \text{supp } P_{Y|X}(\cdot | x) \cap \text{supp } P_{Y|X}(\cdot | x') = \emptyset$ (Bob knows x)

Examples: $\mathbf{P}_{Y|X}^{\text{unimpr}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1/3 & 2/3 \end{pmatrix},$

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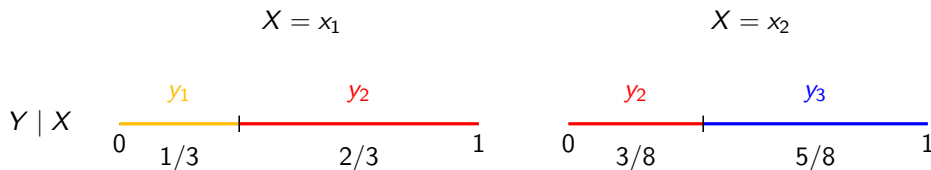
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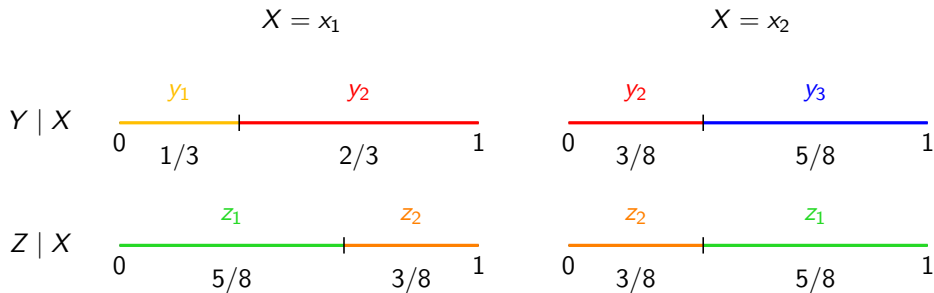
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Unimprovable P_{XY} — illustration of necessity

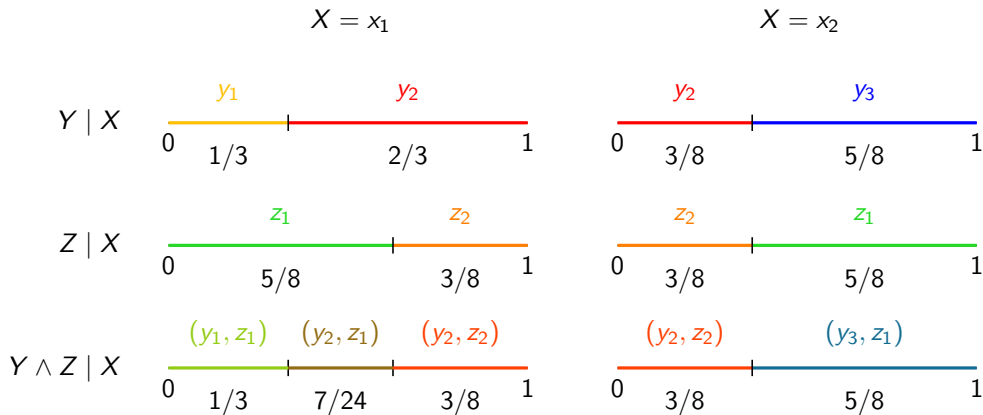


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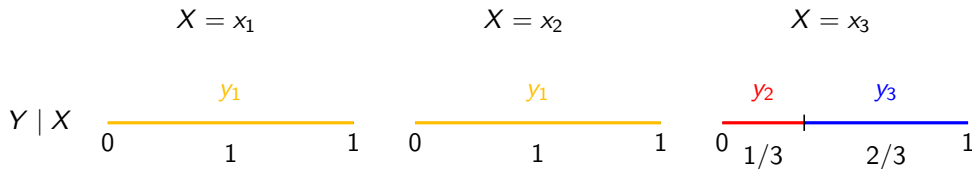
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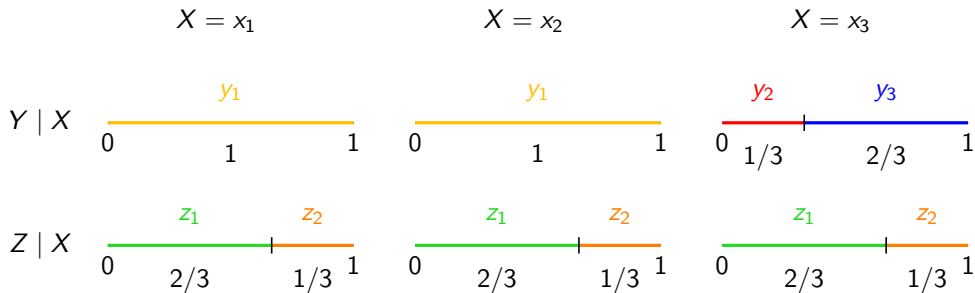


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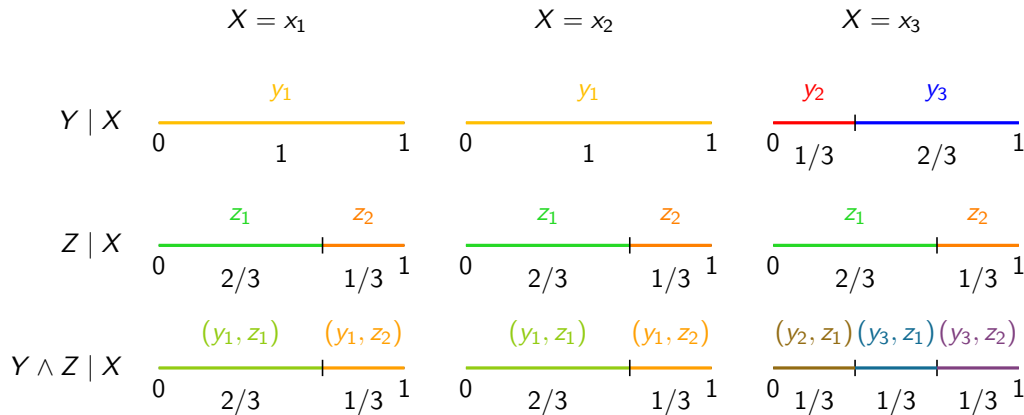


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Toward Blackwell maxima

Say $P_{Y|X}$ is improvable.

How much information about X can Alice reveal to Bob under perfect secrecy?

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Q: What are the Blackwell maximal elements of the implementable set?

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We don't have a complete answer yet. But consider the following exercise.

Alice observes X and Y and announces $Z = z$ according to some Q_{XYZ} .

Now, Alice knows (X, Y, Z) , Bob knows (Y, Z) and Eve knows Z .

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If $Q_{Y|XZ}(\cdot | \cdot, z)$ is improvable, Alice could say more conditional on $Z = z$.

Structure of Blackwell maxima

Proposition 2: If $\mathbf{K}$ is Blackwell maximal among implementable signals, then for any (Z, Q_{XYZ}) that implements $\mathbf{K}$

for all $z \in \text{supp } Q_Z$, $Q_{Y|XZ}(\cdot | \cdot, z)$ is unimprovable.

Structure of Blackwell maxima

Proposition 2: If $\mathbf{K}$ is Blackwell maximal among implementable signals, then for any (Z, Q_{XYZ}) that implements $\mathbf{K}$

for all $z \in \text{supp } Q_Z$, $Q_{Y|XZ}(\cdot | \cdot, z)$ is unimprovable.

Now, suppose $\mathbf{Q}_{YZ|X}$ is Blackwell maximal.

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and therefore is Blackwell-equivalent to a partitional signal.

At Blackwell-maxima,

conditional on each Z -realization, Y is essentially a partitional signal of X .

Z identifies a partition of $\text{supp } P_X$ and Y identifies a cell.

Structure of Blackwell maxima

Corollary: If $Q_{YZ|X}$ is Blackwell maximal among implementable signals, then after any realization $(y, z) \in \text{supp } Q_{YZ}$, Bob's posterior about X satisfies

$$Q_{X|YZ}(x \mid y, z) = \mathbb{1}_A(x) \frac{P_X(x)}{P_X(A)} \quad \text{for some } A \subseteq \text{supp } P_X.$$

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Bob's posterior has the likelihood ratios within A at P_X ,

and excludes $\text{supp } P_X \setminus A$.

Part IV — decision problems under perfect secrecy

Decision problems under perfect secrecy

Say Bob faces a decision problem $D = (A, u)$. $(u : A \times \text{supp } P_X \rightarrow \mathbb{R}, A \text{ finite})$

Alice is aligned with Bob, and can provide information under perfect secrecy.

Decision problems under perfect secrecy

Say Bob faces a decision problem $D = (A, u)$. ($u : A \times \text{supp } P_X \rightarrow \mathbb{R}$, A finite)

Alice is aligned with Bob, and can provide information under perfect secrecy.

Her problem is to choose Q_{XYZ} to maximize

$$\sum_{z=1}^p \sum_{y=1}^m Q_{YZ}(y, z) \left(\max_{a \in A} \sum_{x=1}^n Q_{X|YZ}(x | y, z) u(a, x) \right)$$

subject to $Q_{Z|X}(\cdot | x) = Q_Z$ for all $x \in \text{supp } P_X$,

and $Q_{Y|X} = P_{Y|X}$.

Decision problems under perfect secrecy (cont'd)

Alice's problem is

$$\begin{aligned} \arg \max_{Q_{XYZ}} \quad & \sum_{z=1}^p \sum_{y=1}^m Q_{YZ}(y, z) \left(\max_{a \in A} \sum_{x=1}^n Q_{X|YZ}(x | y, z) u(a, x) \right) \\ \text{s.t.} \quad & Q_{Y|X}(y | x) = P_{Y|X}(y | x) \quad \forall (x, y) \\ & Q_{Z|X}(\cdot | x) = Q_Z \quad \forall x \end{aligned}$$

Decision problems under perfect secrecy (cont'd)

Alice's problem is

$$\arg \max_{Q_{YZ|X}} \sum_{z=1}^p \sum_{y=1}^m \left(\max_{a \in A} \sum_{x=1}^n P_X(x) Q_{YZ|X}(y, z | x) u(a, x) \right)$$

$$\text{s.t.} \quad \sum_{z=1}^p Q_{YZ|X}(y, z | x) = P_{Y|X}(y | x) \quad \forall (x, y)$$

$$Q_{Z|X}(\cdot | x) = Q_Z \quad \forall x$$

Decision problems under perfect secrecy (cont'd)

Alice's problem is

$$\begin{aligned} \arg \max_{Q_{Y|XZ}, Q_{Z|X}} \quad & \sum_{z=1}^p \sum_{y=1}^m \left(\max_{a \in A} \sum_{x=1}^n P_X(x) Q_{Z|X}(z | x) Q_{Y|XZ}(y | x, z) u(a, x) \right) \\ \text{s.t.} \quad & \sum_{z=1}^p Q_{Z|X}(z | x) Q_{Y|XZ}(y | x, z) = P_{Y|X}(y | x) \quad \forall (x, y) \\ & Q_{Z|X}(\cdot | x) = Q_Z \quad \forall x \end{aligned}$$

Decision problems under perfect secrecy (cont'd)

Alice's problem is

$$\begin{aligned} \arg \max_{Q_{Y|XZ}, Q_Z} \quad & \sum_{z=1}^p \sum_{y=1}^m \left(\max_{a \in A} \sum_{x=1}^n P_X(x) Q_Z(z) Q_{Y|XZ}(y | x, z) u(a, x) \right) \\ \text{s.t.} \quad & \sum_{z=1}^p Q_Z(z) Q_{Y|XZ}(\cdot | \cdot, z) = P_{Y|X} \end{aligned}$$

Decision problems under perfect secrecy (cont'd)

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At least one maximizer is Blackwell maximal.

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At Blackwell maxima we can take each $Q_{Y|XZ}(\cdot | \cdot, z) \simeq \mathbf{E}^{(z)}$,

where $\mathbf{E}^{(z)}$ is a partitional signal. ($\mathbf{E}^{(z)}$ has a single positive entry in each row).

Decision problems under perfect secrecy (cont'd)

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$$\begin{aligned} \arg \max_{Q_{Y|XZ}, Q_Z} \quad & \sum_{z=1}^p \sum_{y=1}^m \left(\max_{a \in A} \sum_{x=1}^n P_X(x) Q_Z(z) Q_{Y|XZ}(y | x, z) u(a, x) \right) \\ \text{s.t.} \quad & \sum_{z=1}^p Q_Z(z) Q_{Y|XZ}(\cdot | \cdot, z) = P_{Y|X} \end{aligned}$$

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There are m^n many $n \times m$ partitional signals. Enumerate them $\mathbf{E}^{(1)}, \dots, \mathbf{E}^{(m^n)}$.

Decision problems under perfect secrecy (cont'd)

Alice's problem is

$$\begin{aligned} \arg \max_{Q_Z} \quad & \sum_{z=1}^{m^n} \sum_{y=1}^m \left(\max_{a \in A} \sum_{x=1}^n P_X(x) Q_Z(z) \mathbf{E}^{(z)}[x, y] u(a, x) \right) \\ \text{s.t.} \quad & \sum_{z=1}^{m^n} Q_Z(z) \mathbf{E}^{(z)} = \mathbf{P}_{Y|X} \end{aligned}$$

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Alice's problem is a linear program

Alice's problem reduces to

$$\begin{aligned} \arg \max_{\lambda \in \mathbb{R}_+^{m^n}} \quad & \sum_{z=1}^{m^n} \lambda_z \sum_{y=1}^m \left(\max_{a \in A} \sum_{x=1}^n P_X(x) \mathbf{E}^{(z)}[X, y] u(a, x) \right) \\ \text{s.t.} \quad & \sum_{z=1}^{m^n} \lambda_z \mathbf{E}^{(z)} = \mathbf{P}_{Y|X} \quad \text{and} \quad \sum_{z=1}^{m^n} \lambda_z = 1 \end{aligned}$$

For small n and m the problem is tractable.

There always is a vertex solution.

Generically, the solution is unique.

Thanks

$$\forall y \in \text{supp } P_Y, \quad \sum_{x \in \text{supp } P_X} P_{Y|X}(y | x) \leq 1$$

The One-Time Pad

Suppose $X \in \{0, 1\}^n$ and $Y \in \{0, 1\}^m$ with $n \leq m$,

and that Y is generated uniformly at random independently of X .

Alice shares Y with Bob beforehand.

The ciphertext is $Z = X \oplus Y$, where $\oplus$ denotes bitwise addition modulo 2.

Example: $X = 011$ and $Y = 1011$, then $Z = 1101$.

Bob, knowing Y , can invert Z : $Z \oplus Y = (X \oplus Y) \oplus Y = X$.

Eve cannot: $\forall x \in \text{supp } P_X, (Z \mid X = x) = (X \oplus Y \mid X = x) = x \oplus Y \sim P_Y$.

Continuous distributions (Strack and Yang, 2024)

If $F_{Y|X}(\cdot | x)$ is continuous for $\forall x$, and $F_{Y|X}(y | \cdot)$ is invertible $\forall y$.

Set $Z = F_{Y|X}(Y | X)$.

Bob, who knows y , can invert z by computing $x = (F_{Y|X}(y | \cdot))^{-1}(z)$.

Eve learns nothing: $(Z | X = x) = F_{Y|X}(Y | x) \sim \mathbf{U}[0, 1]$.

Theorem one — necessity sketch

Suppose Q_{XYZ} solves Alice's problem.

Let $\phi(x, y) = \{z \in \text{supp } Z : Q_{XYZ}(x, y, z) > 0\}$.

For any $y \in \text{supp } Y$,

$$\begin{aligned} \sum_{x \in \text{supp } X} \frac{P_{XY}(x, y)}{P_X(x)} &= \sum_{x \in \text{supp } X} \frac{1}{P_X(x)} \sum_{z \in \text{supp } Z} Q_{XYZ}(x, y, z) \\ &= \sum_{x \in \text{supp } X} \frac{1}{P_X(x)} \sum_{z \in \phi(x, y)} Q_{XYZ}(x, y, z) \leq \sum_{x \in \text{supp } X} \frac{1}{P_X(x)} \sum_{z \in \phi(x, y)} Q_{XZ}(x, z) \\ &= \sum_{x \in \text{supp } X} \frac{1}{P_X(x)} \sum_{z \in \phi(x, y)} P_X(x) Q_Z(z) = \sum_{x \in \text{supp } X} \sum_{z \in \phi(x, y)} Q_Z(z) \leq 1. \end{aligned}$$

Theorem one — sufficiency sketch

Suppose that for all $y \in \{1, \dots, m\}$ we have $\sum_{x=1}^n P_{Y|X}(y | x) \leq 1$.

Let $\mathbf{P}_{Y|X}$ be the $n \times m$ matrix whose xy -th entry is $P_{Y|X}(y | x)$.

Then $\mathbf{P}_{Y|X}$ is column-substochastic, with $n \leq m$.

Let $\mathbf{P}_{Y|X}^{\text{ext}}$ be an $m \times m$ doubly-stochastic extension of $\mathbf{P}_{Y|X}$.

By the Birkhoff-von Neumann Theorem, $\mathbf{P}_{Y|X}^{\text{ext}}$ can be expressed as a convex combination of permutation matrices: $\mathbf{P}_{Y|X}^{\text{ext}} = \sum_{z=1}^p \lambda_z \mathbf{\Pi}^{(z)}$.

Define a joint distribution $Q_{XYZ}(x, y, z) = \lambda_z P_X(x) \mathbf{\Pi}^{(z)}[x, y]$.

Theorem two — illustration of sufficiency

Take $\mathbf{P}_X = \begin{pmatrix} 1/4 \\ 3/4 \end{pmatrix}$ and $\mathbf{P}_{Y|X} = \begin{pmatrix} 2/6 & 4/6 \\ 1/6 & 5/6 \end{pmatrix}$. Consider $\mathbf{K} = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}$

Then $L(\mathbf{K}) = \left\{ \begin{pmatrix} 2 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 2 & 0 \\ 0 & 0 \end{pmatrix} \right\} \cup \left\{ \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ \lambda & 2 - \lambda \end{pmatrix} : \lambda \in [0, 2] \right\}$

Choosing $\mathbf{\Theta} = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \\ 0 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} + \frac{1}{6} \begin{pmatrix} 0 & 2 \\ 2 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 2 \end{pmatrix}$

gives $\mathbf{P}_{Y|X} = \mathbf{K}\mathbf{\Theta} = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \\ 0 & 1 \end{pmatrix}$

Theorem two — illustration of sufficiency (cont'd)

The joint that solves the problem is given by $\mathbf{Q}_{XYZ}(\cdot, \cdot, z) = \lambda_z \text{diag}(\mathbf{P}_X) \mathbf{K} \mathbf{\Pi}^{(z)}$

$$\mathbf{Q}_{XYZ}(\cdot, \cdot, z_1) = \frac{1}{3} \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix} \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix}$$

$$\mathbf{Q}_{XYZ}(\cdot, \cdot, z_2) = \frac{1}{6} \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix} \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 2 & 0 \\ 0 & 0 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 0 & 1/4 \\ 3/4 & 0 \end{pmatrix}$$

$$\mathbf{Q}_{XYZ}(\cdot, \cdot, z_3) = \frac{1}{2} \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix} \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 1/4 \\ 0 & 3/4 \end{pmatrix}$$

For informative 2×2 signals, $\text{co}(L(\mathbf{K})) = \text{Birkhoff}_{2 \times 2}$

Suppose $\mathbf{K}$ is a 2×2 row-stochastic matrix with $\mathbf{K}_{11} \neq \mathbf{K}_{21}$.

Any $\mathbf{\Pi} \in L(\mathbf{K})$ is a 2×2 non-negative matrix that (a) has at most one positive element in each column, and (b) satisfies $\mathbf{K}\mathbf{\Pi}\mathbf{1}_2 = \mathbf{1}_2$.

$$\begin{aligned}\text{By (b)} \quad \mathbf{K}_{11}(\mathbf{\Pi}_{11} + \mathbf{\Pi}_{12}) + (1 - \mathbf{K}_{11})(\mathbf{\Pi}_{21} + \mathbf{\Pi}_{22}) &= 1 \\ \mathbf{K}_{21}(\mathbf{\Pi}_{11} + \mathbf{\Pi}_{12}) + (1 - \mathbf{K}_{21})(\mathbf{\Pi}_{21} + \mathbf{\Pi}_{22}) &= 1.\end{aligned}$$

$$\text{Subtracting, } (\mathbf{K}_{11} - \mathbf{K}_{21})(\mathbf{\Pi}_{11} + \mathbf{\Pi}_{12} - \mathbf{\Pi}_{21} - \mathbf{\Pi}_{22}) = 0.$$

$$\text{Since } \mathbf{K}_{11} \neq \mathbf{K}_{21}, \text{ then } \mathbf{\Pi}_{11} + \mathbf{\Pi}_{12} = \mathbf{\Pi}_{21} + \mathbf{\Pi}_{22} = 1.$$

Then, by (a), $\mathbf{\Pi}$ is a permutation of $\mathbf{I}_2$.

Hence, $\text{co}(L(\mathbf{K}))$ is the 2×2 Birkhoff polytope.

Structure of Blackwell maxima — illustration

Consider a joint distribution Q_{XYZ} .

Suppose $Q_{Y|XZ}(\cdot | \cdot, z)$ is unimprovable for all $z \in \text{supp } Q_Z$.

For instance, $\mathbf{Q}_{Y|XZ}(\cdot | \cdot, z_1) = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{Q}_{Y|XZ}(\cdot | \cdot, z_2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$.

By perfect secrecy, $Q_{YZ|X}(y, z | x) = Q_Z(z) Q_{Y|XZ}(y | x, z)$.

And therefore $\mathbf{Q}_{YZ|X} = \left(\begin{array}{cc|cc} (y_1, z_1) & (y_2, z_1) & (y_1, z_2) & (y_2, z_2) \\ Q_Z(z_1) & 0 & Q_Z(z_2) & 0 \\ Q_Z(z_1) & 0 & 0 & Q_Z(z_2) \\ 0 & Q_Z(z_1) & 0 & Q_Z(z_2) \end{array} \right)$.

Structure of Blackwell maxima — proof sketch

Fix $(y, z) \in \text{supp } Q_{YZ}$.

Suppose that $Q_{Y|XZ}(\cdot | \cdot, z)$ is unimprovable,

and that $Q_{YZ|X}(y, z | x_0) \neq Q_{YZ|X}(y, z | x_1) > 0$ for some $x_0 \neq x_1$

By perfect secrecy, $Q_{YZ|X}(y, z | x) = Q_Z(z)Q_{Y|XZ}(y | x, z)$

and therefore, $Q_{Y|XZ}(y | x_0, z) \neq Q_{Y|XZ}(y | x_1, z) > 0$.

Then $Q_{Y|XZ}(\cdot | x_0, z) \cap Q_{Y|XZ}(\cdot | x_1, z)$ is not empty (it contains y).

Since $Q_{Y|XZ}(\cdot | \cdot, z)$ is unimprovable, $Q_{Y|XZ}(y | x_0, z) = Q_{Y|XZ}(y | x_1, z) = 1$.